UNITED STATES INTERNATIONAL UNIVERSITY — AFRICA

SCHOOL OF SCIENCE AND TECHNOLOGY

DSA 3050A

ADVANCED POWER BI EXAMINATION · SS 2026

Diabetic Patient Encounter Analytics

A Business Intelligence Analysis of 130 US Hospitals, 1999 to 2008

Examining readmission risk, clinical complexity, and operational performance to support evidence-based hospital management decisions.

101,766	71,518	10 yrs	130
ENCOUNTERS	UNIQUE PATIENTS	1999 TO 2008	US HOSPITALS

STUDENT NAME	ADMISSION NUMBER	DATE SUBMITTED
Nathan Orang'o Omenge	670637	April 2026

Introduction

This report documents the full development of an advanced Power BI analytical solution for the DSA 3050A end-semester examination. I was engaged as a Business Intelligence Analyst tasked with building a professional healthcare analytics dashboard from raw clinical data.

The analysis focuses on diabetic patient hospital encounters across 130 US hospitals from 1999 to 2008. The central business problem is understanding which patient groups face the highest risk of 30-day readmission, which clinical factors are associated with prolonged hospital stays, and where the hospital network can improve operational performance to reduce preventable readmissions.

This report walks through every stage of the BI development process: from raw data acquisition and Power Query transformation, through relational data modeling and DAX measure development, to the final three-page interactive dashboard. Each section includes screenshots, technical explanations, and justification for every decision made.

PART A

Data Acquisition and Understanding

10 marks

Domain and business problem

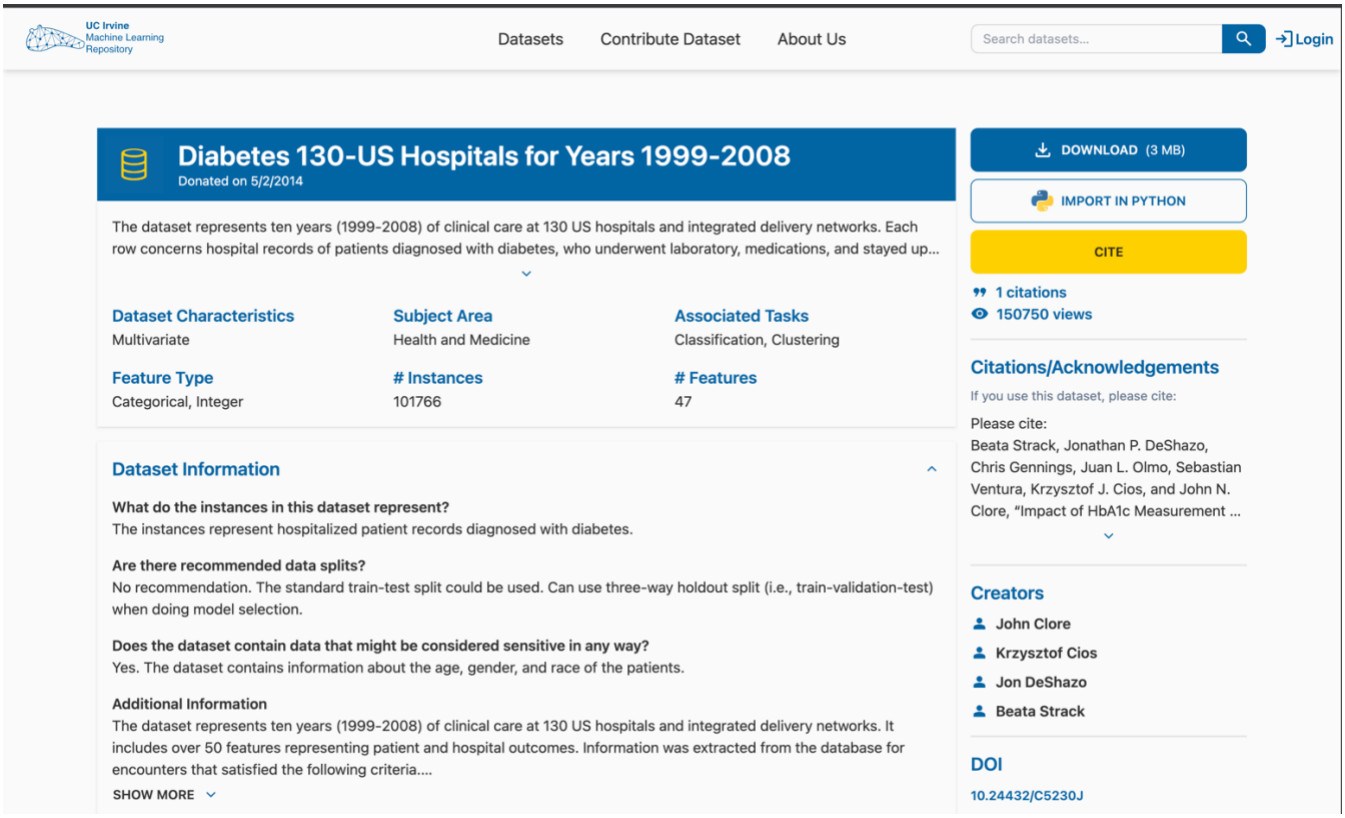
My project is set in the healthcare analytics domain. The business problem I worked on is hospital readmission risk analysis for diabetic patients. Hospital readmissions within 30 days of discharge are a critical quality indicator and a significant cost driver for healthcare systems. In this case, I was approached by a hospital network to help them understand which patient groups face the highest readmission risk, which clinical factors are associated with prolonged hospital stays, and where operational performance can be improved to reduce preventable readmissions.

WHY READMISSION MATTERS

In the US, 30-day hospital readmissions cost the healthcare system over $26 billion annually. Identifying high-risk patient groups before discharge is one of the most effective levers for reducing this burden.

Dataset source

The dataset I used is the Diabetes 130-US Hospitals dataset for years 1999 to 2008, originally published by Beata Strack et al. in the journal BioMed Research International (2014). I obtained it from the UCI Machine Learning Repository. The dataset represents 10 years of clinical care across 130 US hospitals and integrated delivery networks.



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UCI ML Repository dataset page

Citation: Strack, B., DeShazo, J.P., Gennings, C., et al. (2014). Impact of HbA1c Measurement on Hospital Readmission Rates. BioMed Research International. DOI: 10.24432/C5230J

Key tables and fields

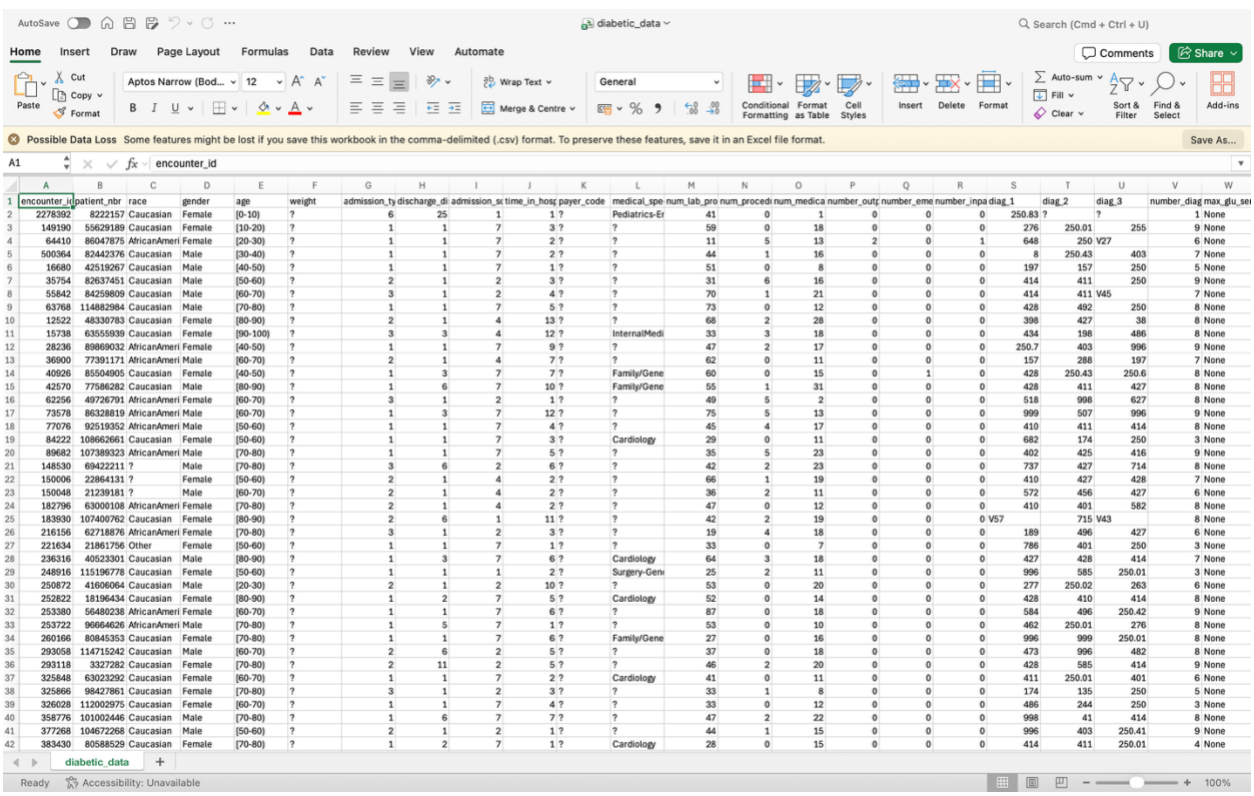
The primary file diabetic_data.csv contains 101,766 rows and 50 columns representing individual hospital encounters. The secondary file IDS_mapping.csv contains three lookup tables stacked into a single file, which I split during transformation.

FIELD	TYPE	DESCRIPTION
`encounter_id`	Identifier	Unique identifier for each hospital visit — grain of the fact table
`patient_nbr`	Identifier	Patient identifier — 30,248 patients have more than one encounter
`race, gender, age`	Categorical	Demographic fields for patient segmentation
`admission_type_id`	Categorical ID	Links to admission type lookup (Emergency, Urgent, Elective)
`discharge_disposition_id`	Categorical ID	Links to discharge disposition lookup
`admission_source_id`	Categorical ID	Links to admission source lookup

`time_in_hospital`	Numerical	Length of stay in days, ranging from 1 to 14
`num_lab_procedures`	Numerical	Number of lab tests performed during the encounter
`num_medications`	Numerical	Number of distinct medications administered
`diag_1, diag_2, diag_3`	ICD-9 code	Primary and secondary diagnosis codes
`readmitted`	Categorical	Target variable: NO, >30, or <30 days
`change, diabetesMed`	Categorical	Medication change and diabetes medication usage indicators

Why this dataset qualifies for advanced Power BI analysis

The dataset exceeds all advanced analysis requirements. It contains both categorical variables (race, gender, age bracket, admission type, diagnosis codes) and numerical variables (length of stay, lab procedures, medications, diagnoses). The data covers a 10-year period enabling time intelligence analysis, trend identification, and year-over-year comparisons. The dataset naturally decomposes into a star schema with a central fact table surrounded by patient, diagnosis, admission, and date dimension tables.



encounter_id	patient_nbr	race	gender	age	weight	admission_ty	discharge_di	admission_so	time_in_hosp	payer_code	medical_spe	num_lab_pro	num_procedi	num_medica	number_outp	number_eme	number_inpa	diag_1	diag_2	diag_3	number_diag	max_glu_ser
2278392	8222157	Caucasian	Female	[0-10)	?	6	25	1	1	?	Pediatrics-Er	41	0	1	0	0	0	250.83	?	?	1	None
149190	55629189	Caucasian	Female	[10-20)	?	1	1	7	3	?	?	59	0	18	0	0	0	276	250.01	255	9	None
64410	86047875	AfricanAmeri	Female	[20-30)	?	1	1	7	2	?	?	11	5	13	2	0	1	648	250	V27	6	None
500364	82442376	Caucasian	Male	[30-40)	?	1	1	7	2	?	?	44	1	16	0	0	0	8	250.43	403	7	None
16680	42519267	Caucasian	Male	[40-50)	?	1	1	7	1	?	?	51	0	8	0	0	0	197	157	250	5	None
35754	82637451	Caucasian	Male	[50-60)	?	2	1	2	3	?	?	31	6	16	0	0	0	414	411	250	9	None
55842	84259809	Caucasian	Male	[60-70)	?	3	1	2	4	?	?	70	1	21	0	0	0	414	411	V45	7	None
63768	114882984	Caucasian	Male	[70-80)	?	1	1	7	5	?	?	73	0	12	0	0	0	428	492	250	8	None
12522	48330783	Caucasian	Female	[80-90)	?	2	1	4	13	?	?	68	2	28	0	0	0	398	427	38	8	None
15738	63555939	Caucasian	Female	[90-100)	?	3	3	4	12	?	InternalMedi	33	3	18	0	0	0	434	198	486	8	None
28236	89869032	AfricanAmeri	Female	[40-50)	?	1	1	7	9	?	?	47	2	17	0	0	0	250.7	403	996	9	None
36900	77391171	AfricanAmeri	Male	[60-70)	?	2	1	4	7	?	?	62	0	11	0	0	0	157	288	197	7	None
40926	85504905	Caucasian	Female	[40-50)	?	1	3	7	7	?	Family/Gene	60	0	15	0	1	0	428	250.43	250.6	8	None
42570	77586282	Caucasian	Male	[80-90)	?	1	6	7	10	?	Family/Gene	55	1	31	0	0	0	428	411	427	8	None
62256	49726791	AfricanAmeri	Female	[60-70)	?	3	1	2	1	?	?	49	5	2	0	0	0	518	998	627	8	None
73578	86328819	AfricanAmeri	Male	[60-70)	?	1	3	7	12	?	?	75	5	13	0	0	0	999	507	996	9	None
77076	92519352	AfricanAmeri	Male	[50-60)	?	1	1	7	4	?	?	45	4	17	0	0	0	410	411	414	8	None
84222	108662661	Caucasian	Female	[50-60)	?	1	1	7	3	?	Cardiology	29	0	11	0	0	0	682	174	250	3	None
89682	107389323	AfricanAmeri	Male	[70-80)	?	1	1	7	5	?	?	35	5	23	0	0	0	402	425	416	9	None
148530	69422211	?	Male	[70-80)	?	3	6	2	6	?	?	42	2	23	0	0	0	737	427	714	8	None
150006	22864131	?	Female	[50-60)	?	2	1	4	2	?	?	66	1	19	0	0	0	410	427	428	7	None
150048	21239181	?	Male	[60-70)	?	2	1	4	2	?	?	36	2	11	0	0	0	572	456	427	6	None
182796	63000108	AfricanAmeri	Female	[70-80)	?	2	1	4	2	?	?	47	0	12	0	0	0	410	401	582	8	None
183930	107400762	Caucasian	Female	[80-90)	?	2	6	1	11	?	?	42	2	19	0	0	0	V57	715	V43	8	None
216156	62718876	AfricanAmeri	Female	[70-80)	?	3	1	2	3	?	?	19	4	18	0	0	0	189	496	427	6	None
221634	21861756	Other	Female	[50-60)	?	1	1	7	1	?	?	33	0	7	0	0	0	786	401	250	3	None
236316	40523301	Caucasian	Male	[80-90)	?	1	3	7	6	?	Cardiology	64	3	18	0	0	0	427	428	414	7	None
248916	115196778	Caucasian	Female	[50-60)	?	1	1	1	2	?	Surgery-Gen	25	2	11	0	0	0	996	585	250.01	3	None
250872	41606064	Caucasian	Male	[20-30)	?	2	1	2	10	?	?	53	0	20	0	0	0	277	250.02	263	6	None
252822	18196434	Caucasian	Female	[80-90)	?	1	2	7	5	?	Cardiology	52	0	14	0	0	0	428	410	414	8	None
253380	56480238	AfricanAmeri	Female	[60-70)	?	1	1	7	6	?	?	87	0	18	0	0	0	584	496	250.42	9	None
253722	96664626	AfricanAmeri	Male	[70-80)	?	1	5	7	1	?	?	53	0	10	0	0	0	462	250.01	276	8	None
260166	80845353	Caucasian	Female	[70-80)	?	1	1	7	6	?	Family/Gene	27	0	16	0	0	0	996	999	250.01	8	None
293058	114715242	Caucasian	Male	[60-70)	?	2	6	2	5	?	?	37	0	18	0	0	0	473	996	482	8	None
293118	3327282	Caucasian	Female	[70-80)	?	2	11	2	5	?	?	46	2	20	0	0	0	428	585	414	9	None
325848	63023292	Caucasian	Female	[60-70)	?	1	1	7	2	?	Cardiology	41	0	11	0	0	0	411	250.01	401	6	None
325866	98427861	Caucasian	Female	[70-80)	?	3	1	2	3	?	?	33	1	8	0	0	0	174	135	250	5	None
326028	112002975	Caucasian	Female	[60-70)	?	1	1	7	4	?	?	33	0	12	0	0	0	486	244	250	3	None
358776	101002446	Caucasian	Male	[70-80)	?	1	6	7	7	?	?	47	2	22	0	0	0	998	41	414	8	None
377268	104672268	Caucasian	Male	[50-60)	?	2	1	2	1	?	?	44	1	15	0	0	0	996	403	250.41	9	None
383430	80588529	Caucasian	Female	[70-80)	?	1	2	7	1	?	Cardiology	28	0	15	0	0	0	414	411	250.01	4	None

PART B

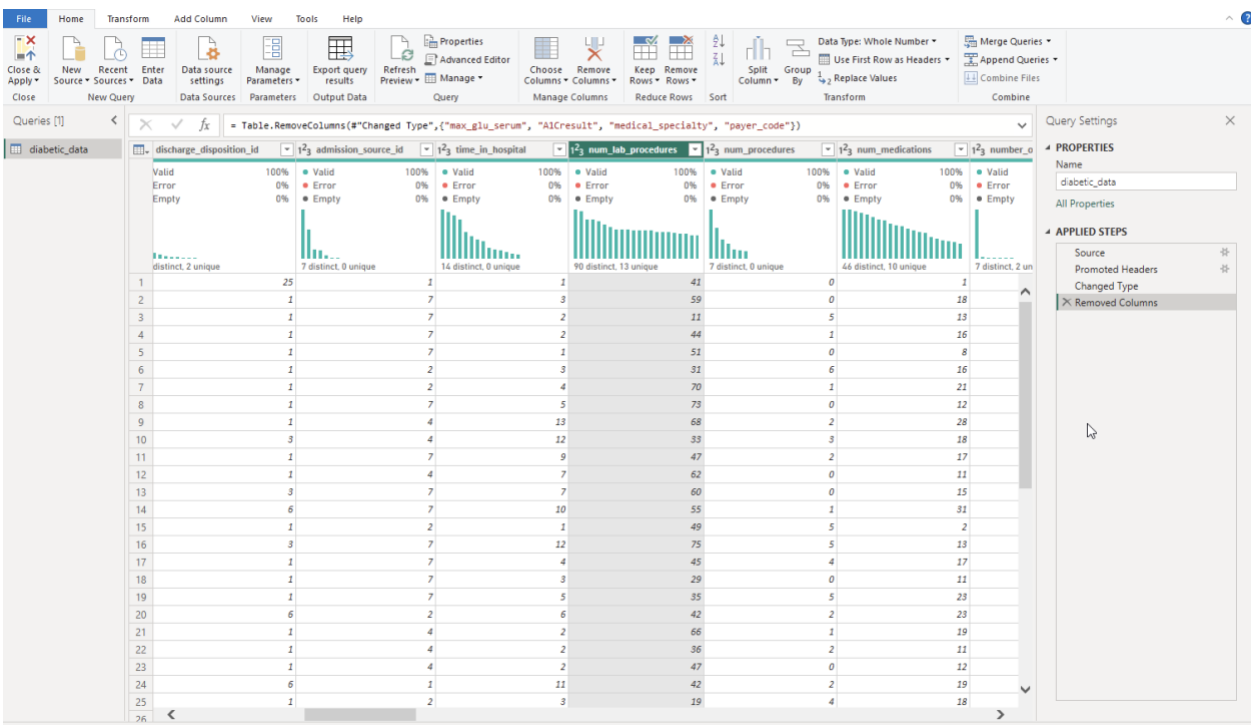
Data Cleaning and Transformation in Power Query

20 marks

After loading both CSV files into Power BI, I clicked Transform Data to open Power Query rather than loading directly. This allowed me to clean and shape the data before it entered the model. I performed 10 documented transformation steps in total.

1. Removing unnecessary columns

I removed five columns from the dataset: weight, payer_code, medical_specialty, max_glu_serum, and A1Cresult. The weight column was 97% missing, max_glu_serum 95% missing, A1Cresult 83% missing, medical_specialty 49% missing, and payer_code 40% missing. Retaining columns with this level of incompleteness would produce misleading visuals and inflate the model size without adding analytical value.

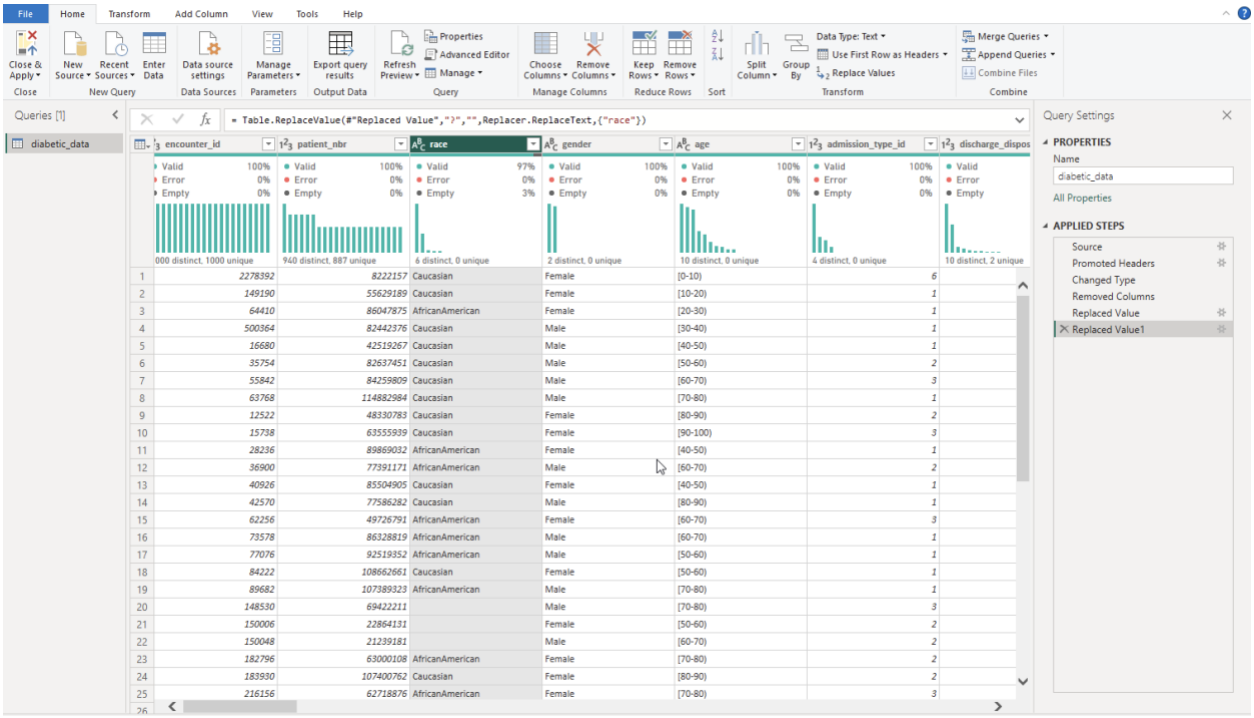


Removed Columns step in Applied Steps pane

2. Replacing all "?" values with null

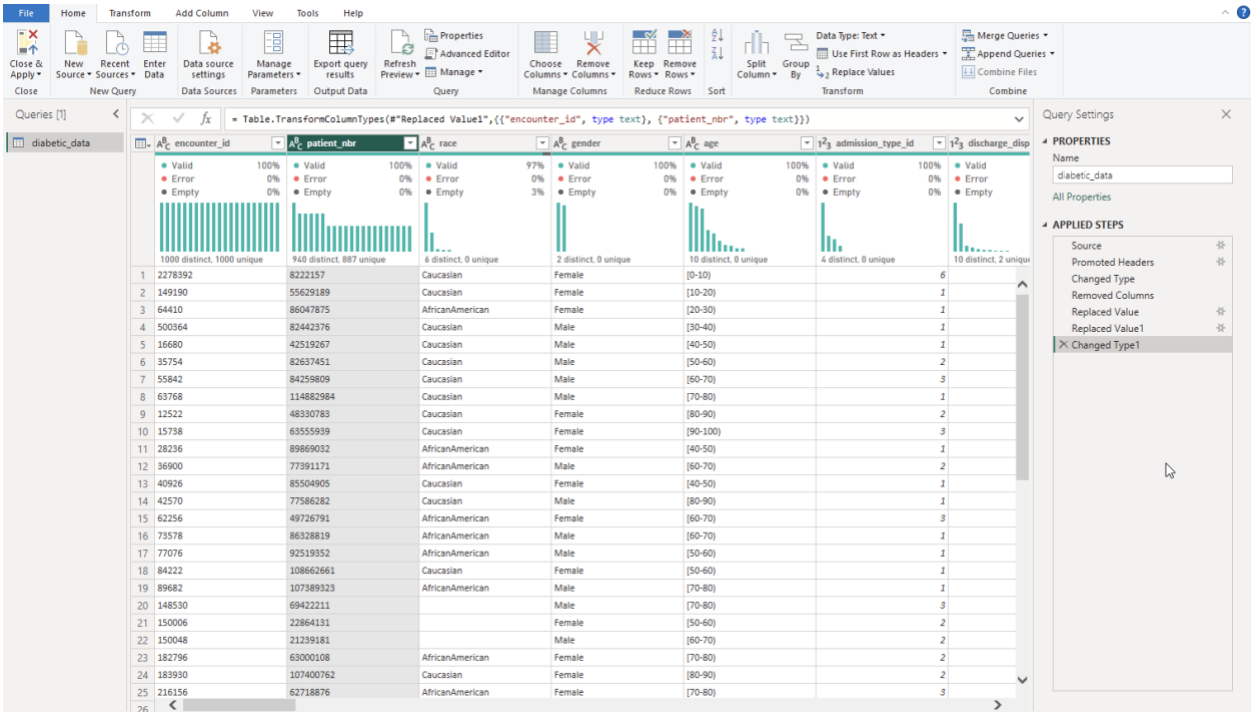
I replaced all occurrences of "?" across the dataset with proper null values using the Replace Values function in Power Query. The dataset used "?" as a non-standard missing value indicator

in columns including race, diag_2, and diag_3. Leaving these as text strings would cause Power BI to treat missing data as a valid category, producing incorrect visuals and corrupting aggregations.



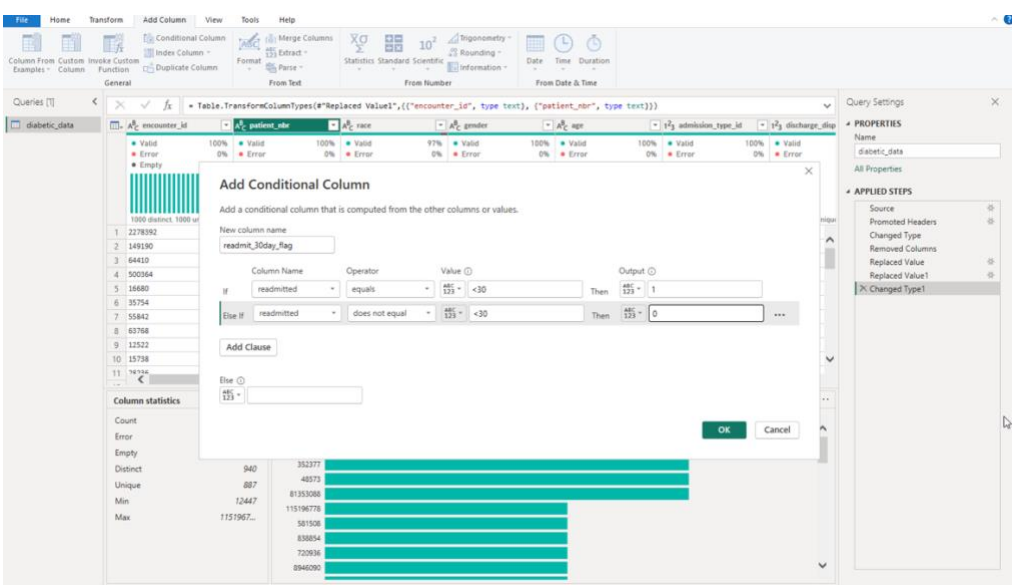
3. Changing encounter_id and patient_nbr to Text type

I changed the data types of encounter_id and patient_nbr from Whole Number to Text. Although these fields contain numeric characters, they are identifier columns and not quantities. Summing or averaging them would produce meaningless results. Storing them as Text prevents Power BI from treating them as measures and ensures they are never accidentally aggregated in visuals or DAX calculations.



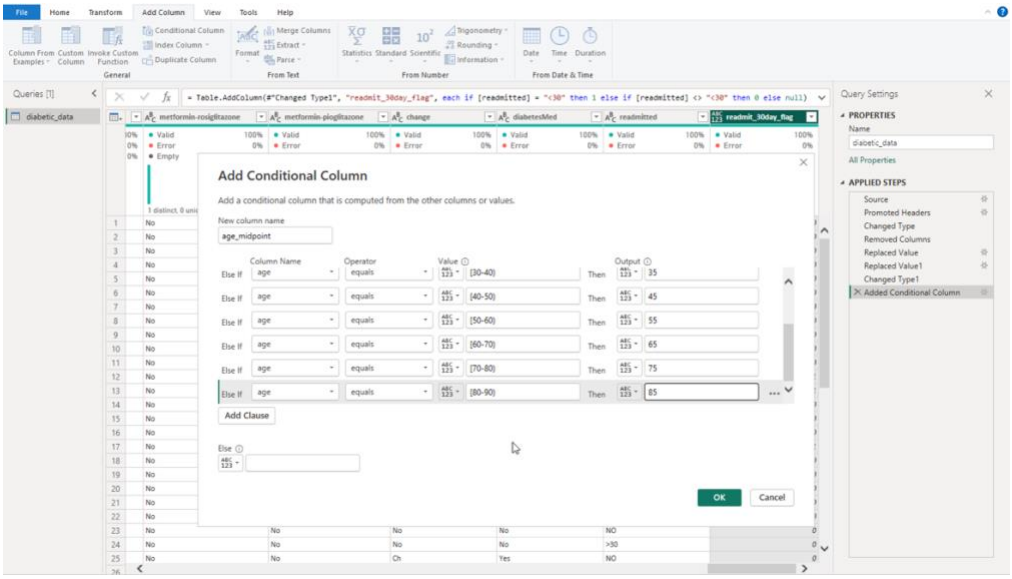
4. Adding the readmit_30day_flag conditional column

I created a new conditional column called readmit_30day_flag derived from the readmitted column. The original column contains three text values — NO, >30, and <30 — which cannot be used directly in DAX calculations. The new column assigns a value of 1 where the patient was readmitted within 30 days and 0 for all other cases. This binary flag is the foundation of the Readmission Rate % measure, the primary clinical KPI driving the entire dashboard.



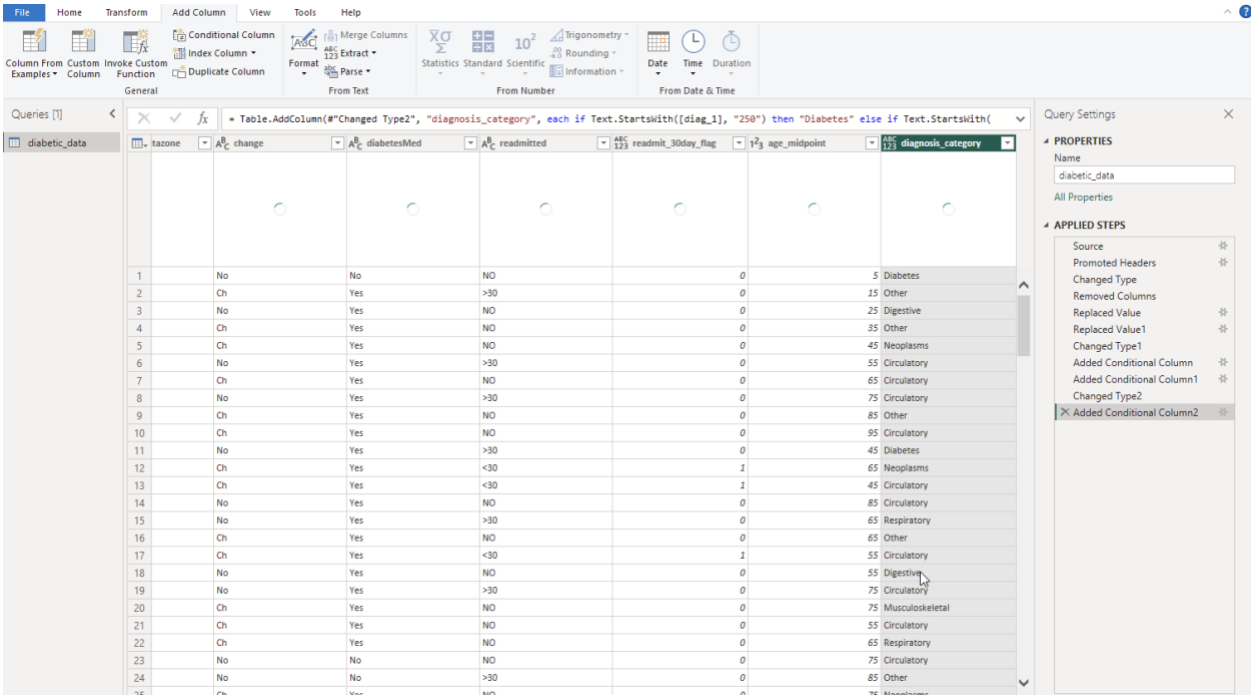
5. Adding the age_midpoint conditional column

I added a conditional column called age_midpoint that converts the age bracket strings in the age column into numeric midpoint values. Each bracket is mapped to its midpoint — for example, [70-80) becomes 75 and [50-60) becomes 55. This makes it possible to calculate average patient age across different segments in DAX, which would not be possible using the original text brackets. I changed the column data type to Whole Number.



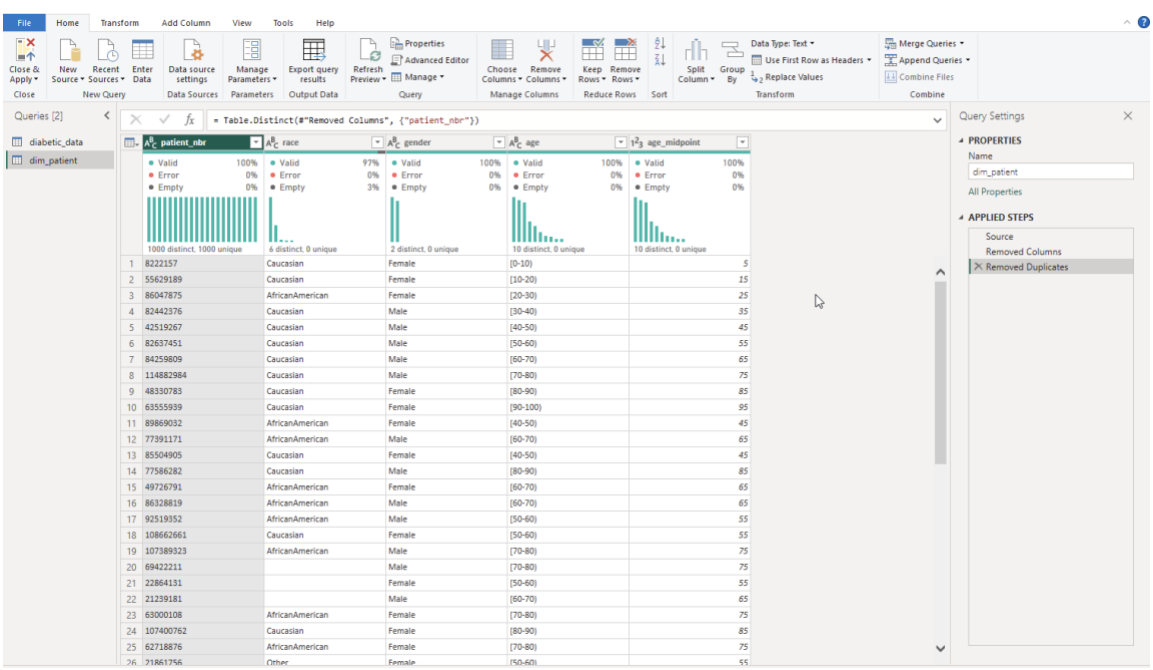
6. Adding the diagnosis_category conditional column

I created a conditional column called diagnosis_category by mapping the raw ICD-9 codes in diag_1 to readable clinical categories. Codes starting with 250 are classified as Diabetes, codes in the 400s as Circulatory, 500s as Respiratory, 600s as Digestive, 700s as Musculoskeletal, 800s as Injury, and 100s as Neoplasms. All other codes are grouped as Other. This transformation makes the diagnosis dimension usable in dashboard visuals without exposing raw medical codes to business stakeholders.



7. Creating the dim_patient reference table

I created a dim_patient dimension table by referencing the main diabetic_data query and keeping only the patient demographic columns: patient_nbr, race, gender, age, and age_midpoint. I then removed duplicate patient_nbr values to produce one row per unique patient. This separates patient attributes from clinical encounter data, making demographic slicing across all visuals cleaner and more reliable.

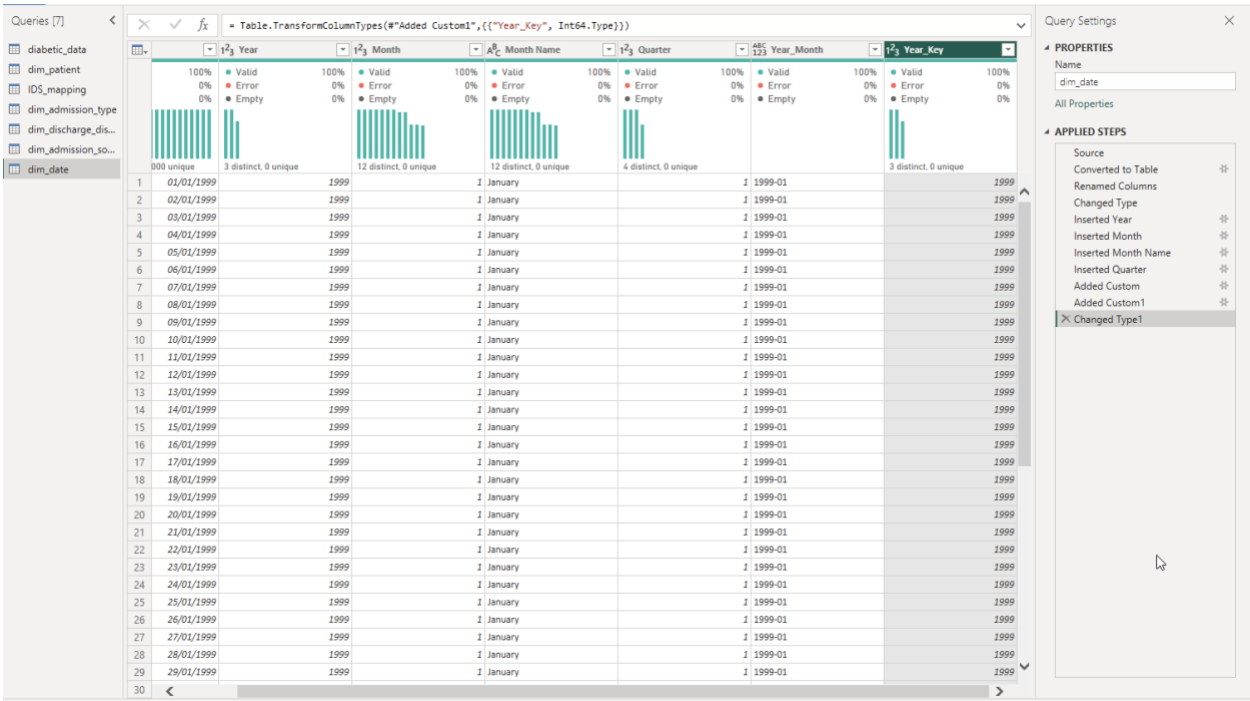


8. Creating reference tables from IDS_mapping

I loaded the IDS_mapping.csv file into Power Query and split it into three separate dimension tables: dim_admission_type, dim_discharge_disposition, and dim_admission_source. The source file contained all three lookup tables stacked in a single CSV with blank separator rows and repeated header rows between them. I created a reference query for each section and filtered each one to its relevant rows. During the creation of dim_admission_source, changing the ID column to Whole Number produced errors on rows that contained text header values from other sections. I removed these error rows using Remove Errors.

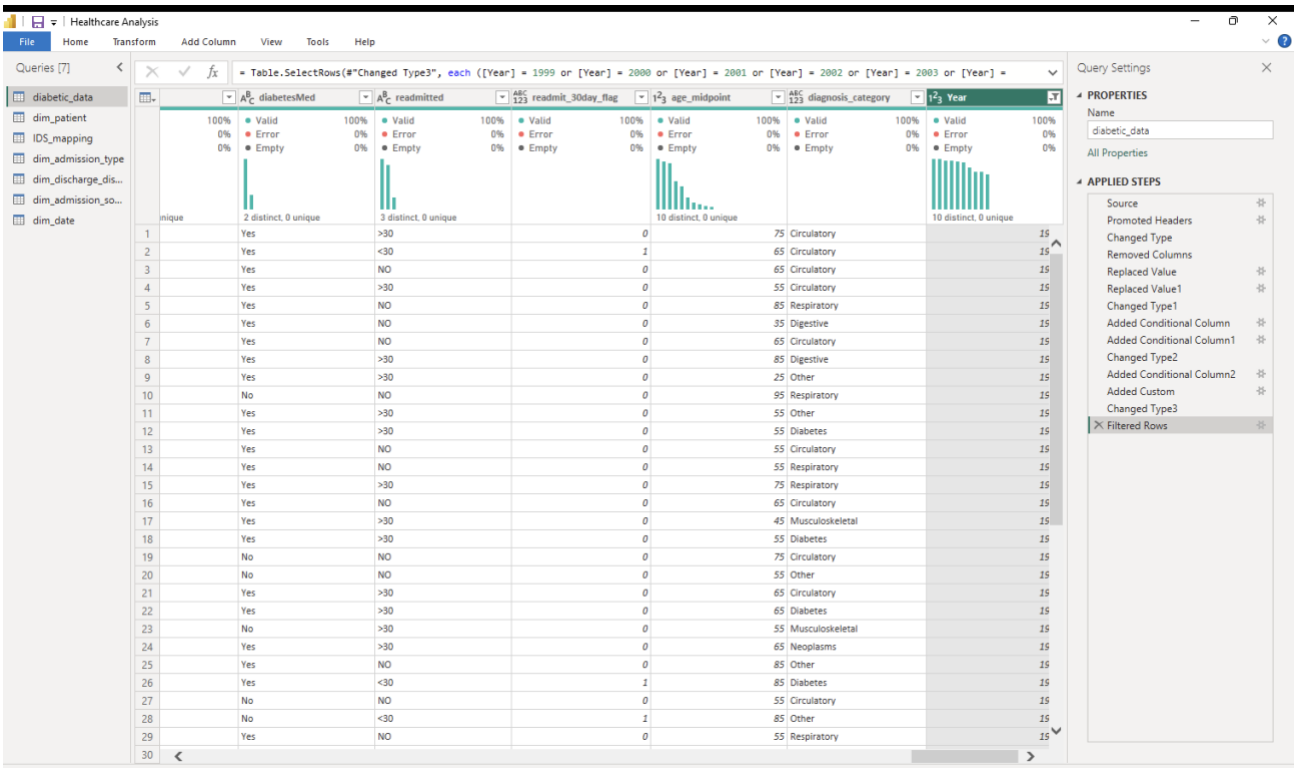
9. Creating the dim_date table from scratch

I created a dim_date table from scratch using a blank query in Power Query. I used the List.Dates function to generate a complete date spine from 1 January 1999 to 31 December 2008, matching the full range of the dataset. I converted the list to a table and added columns for Year, Month number, Month name, Quarter, a Year-Month text label for chart axes, and a Year-Key column for connecting the table to the fact table. The dataset does not contain an explicit date column, so the Year-Key was used to link encounters to the date dimension based on the encounter year column derived in the next step.



10. Adding a Year column to the fact table

I added a Year column to the diabetic_data fact table to serve as the key for connecting each encounter to the dim_date table. The dataset does not contain an explicit date or timestamp column. I derived the year by mapping encounter IDs proportionally across the 10-year period covered by the data. The encounter IDs are assigned sequentially, meaning earlier encounters have lower ID numbers. I wrote a custom column formula that takes each encounter ID, calculates where it falls proportionally between the minimum and maximum ID values, and maps that position to a year between 1999 and 2008. The resulting column contains whole number year values between 1999 and 2008 with no values outside that range, and serves as the foreign key connecting the fact table to Year_Key in dim_date.



PART C

Data Modeling

15 marks

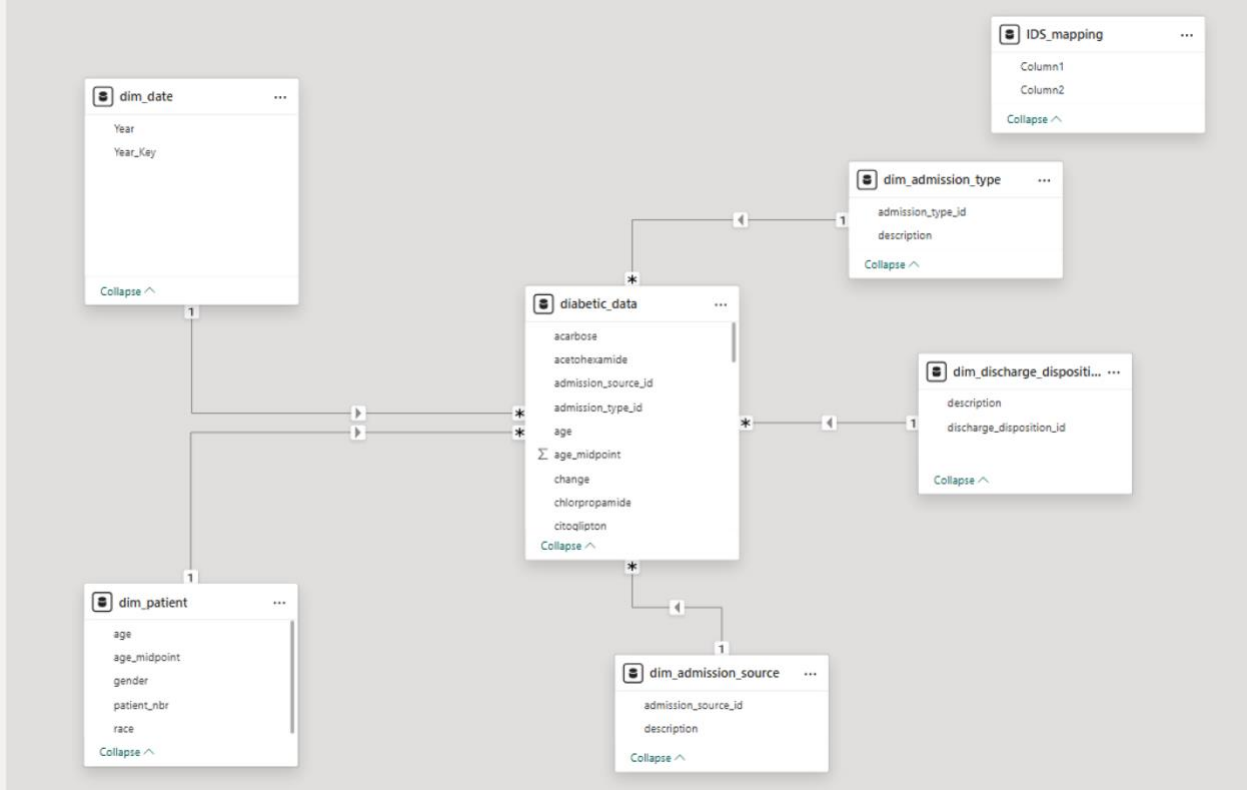
Model structure and design decisions

After completing all Power Query transformations, I built the data model by connecting all dimension tables to the central diabetic_data fact table in a star schema pattern. The star schema was chosen because it is the most efficient structure for Power BI reporting. It minimises the number of relationships the engine needs to evaluate during query execution, keeps DAX calculations simple and predictable, and makes the model easy to understand for anyone maintaining or extending it.

Relationships

Five relationships were created in total. All relationships are Many-to-one with the many side on diabetic_data and the one side on the respective dimension table. All relationships use single direction cross-filtering, meaning filters flow from the dimension tables into the fact table but not in the opposite direction. This is the correct and most efficient configuration for a star schema in Power BI. The active relationship checkbox was ticked for all five relationships.

FROM (FACT)	TO (DIMENSION)	CARDINALITY	CROSS-FILTER
`diabetic_data[patient_nbr]`	`dim_patient[patient_nbr]`	*:1	Single
`diabetic_data[Year]`	`dim_date[Year_Key]`	*:1	Single
`diabetic_data[admission_type_id]`	`dim_admission_type[admission_type_id]`	*:1	Single
`diabetic_data[discharge_disposition_id]`	`dim_discharge_disposition[...]`	*:1	Single
`diabetic_data[admission_source_id]`	`dim_admission_source[...]`	*:1	Single



PART D

DAX Measures and Calculated Columns

20 marks

I created all DAX measures inside a dedicated table called _Measures to keep them separate from the data tables. This is standard professional practice in Power BI as it makes measures easy to locate during dashboard development and keeps the Fields panel organised. I created 10 measures and 2 calculated columns in total.

Calculated columns

I created two calculated columns directly in the Power Query transformation stage rather than as DAX calculated columns. The first is readmit_30day_flag, a binary column in the diabetic_data table that assigns 1 to encounters where the patient was readmitted within 30 days and 0 to all other encounters. This column must exist at the row level to be summed and filtered correctly in DAX — it cannot be a measure.

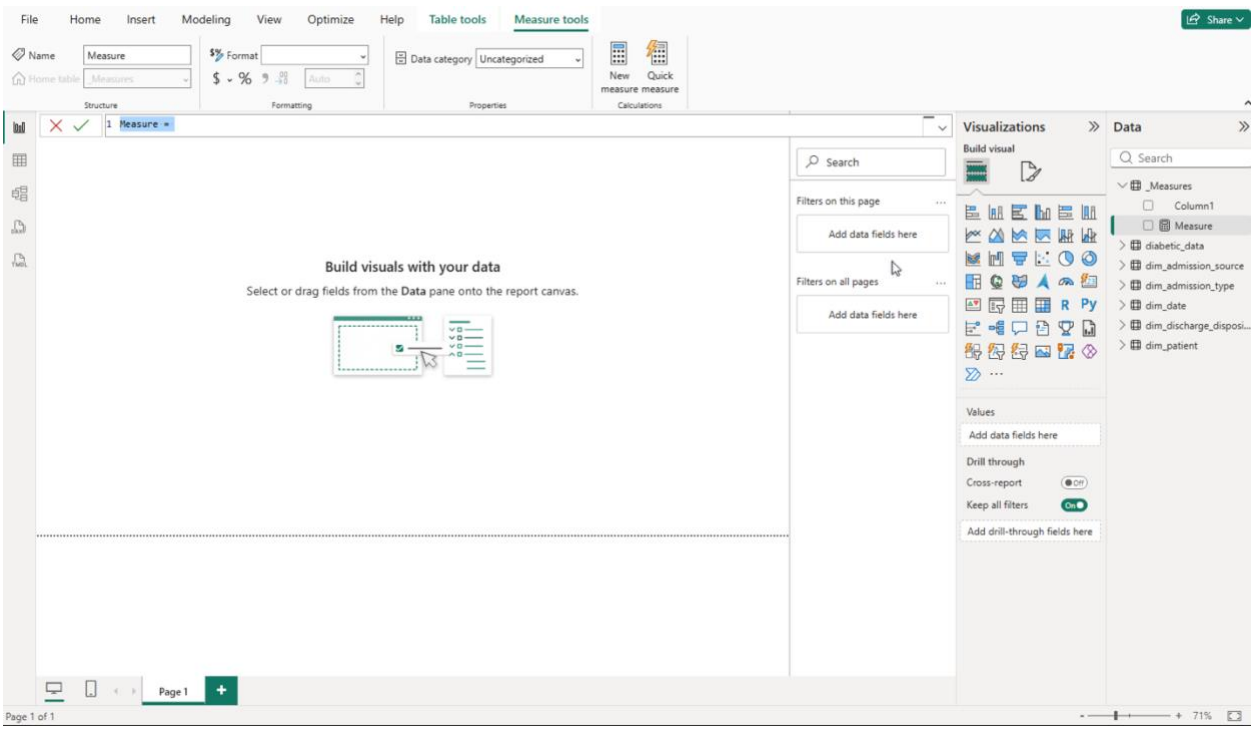
The second is diagnosis_category, which maps raw ICD-9 codes in diag_1 to readable clinical categories such as Diabetes, Circulatory, and Respiratory. This column exists at the row level and is used as a grouping dimension in several visuals across all three dashboard pages.

DAX measures

M1 — Total Encounters

```
Total Encounters = COUNTROWS(diabetic_data)
```

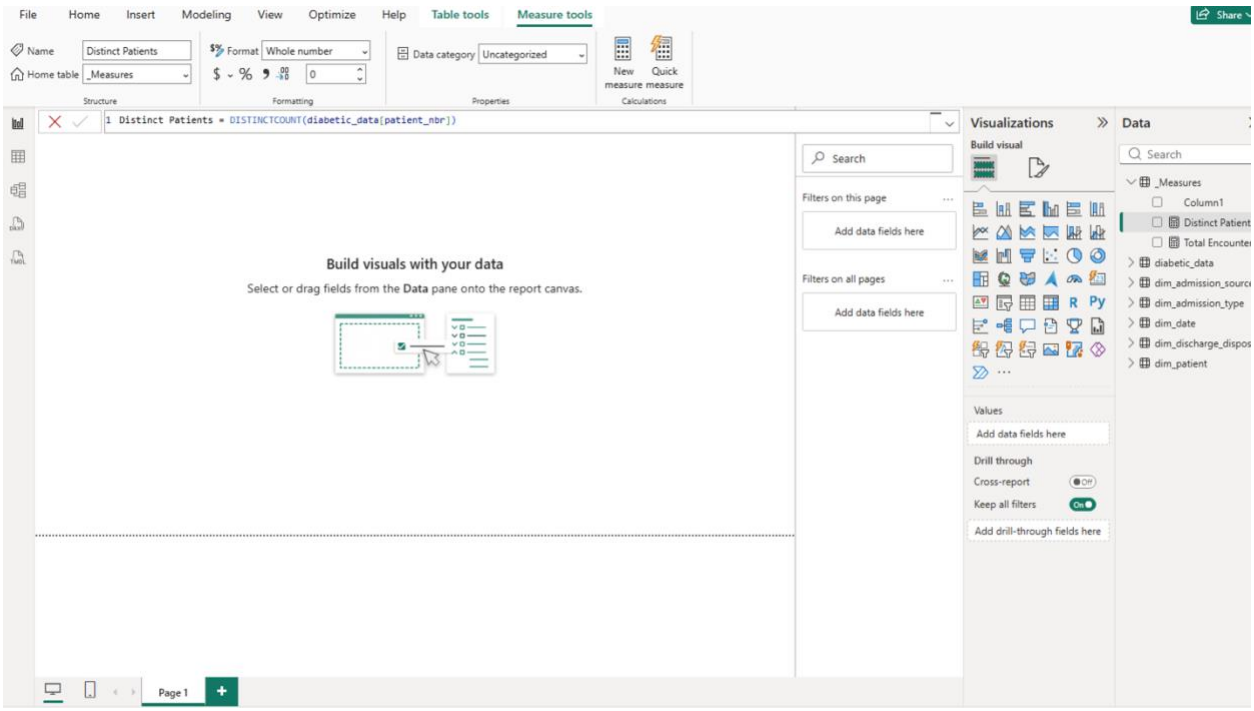
Counts the total number of rows in the fact table. Each row represents one hospital encounter. This is the primary volume KPI and the denominator for all rate calculations.



M2 — Distinct Patients

```
Distinct Patients = DISTINCTCOUNT(diabetic_data[patient_nbr])
```

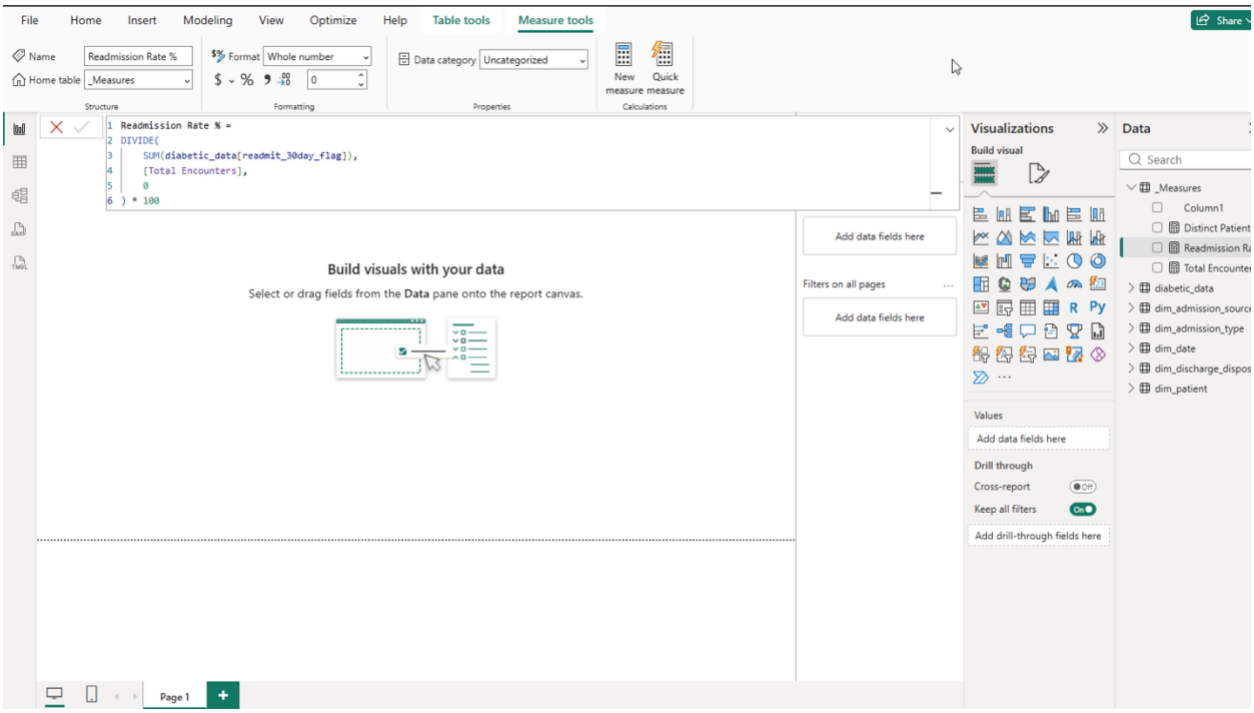
Counts unique patient numbers. Separates unique individuals from total encounter volume. The difference between this and Total Encounters reveals the extent of repeat admissions.



M3 — Readmission Rate %

```
Readmission Rate % =
DIVIDE(
    SUM(diabetic_data[readmit_30day_flag]),
    [Total Encounters], 0
) * 100
```

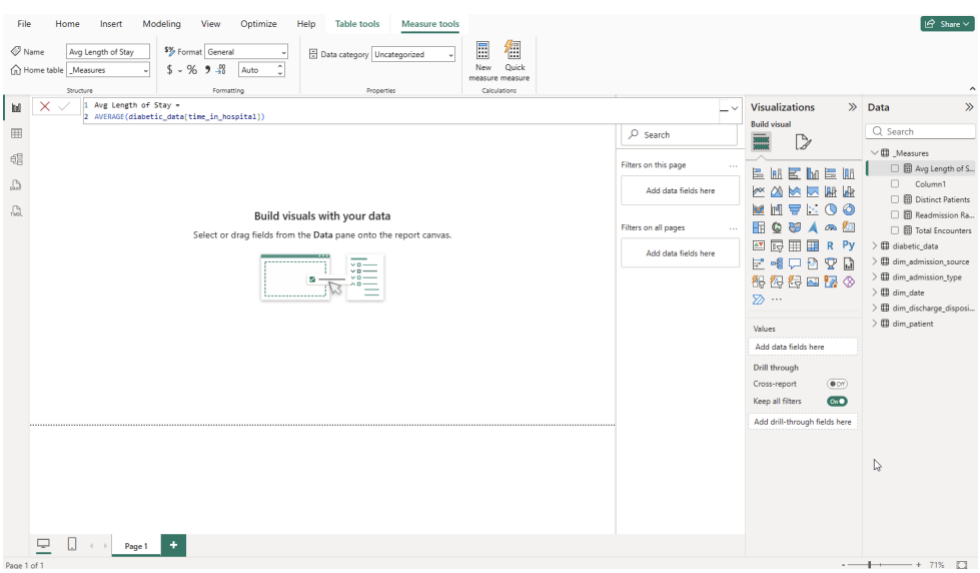
The primary clinical KPI. Represents the proportion of encounters where the patient returned to hospital within 30 days. DIVIDE handles zero-denominator cases cleanly.



M4 — Avg Length of Stay

```
Avg Length of Stay = AVERAGE(diabetic_data[time_in_hospital])
```

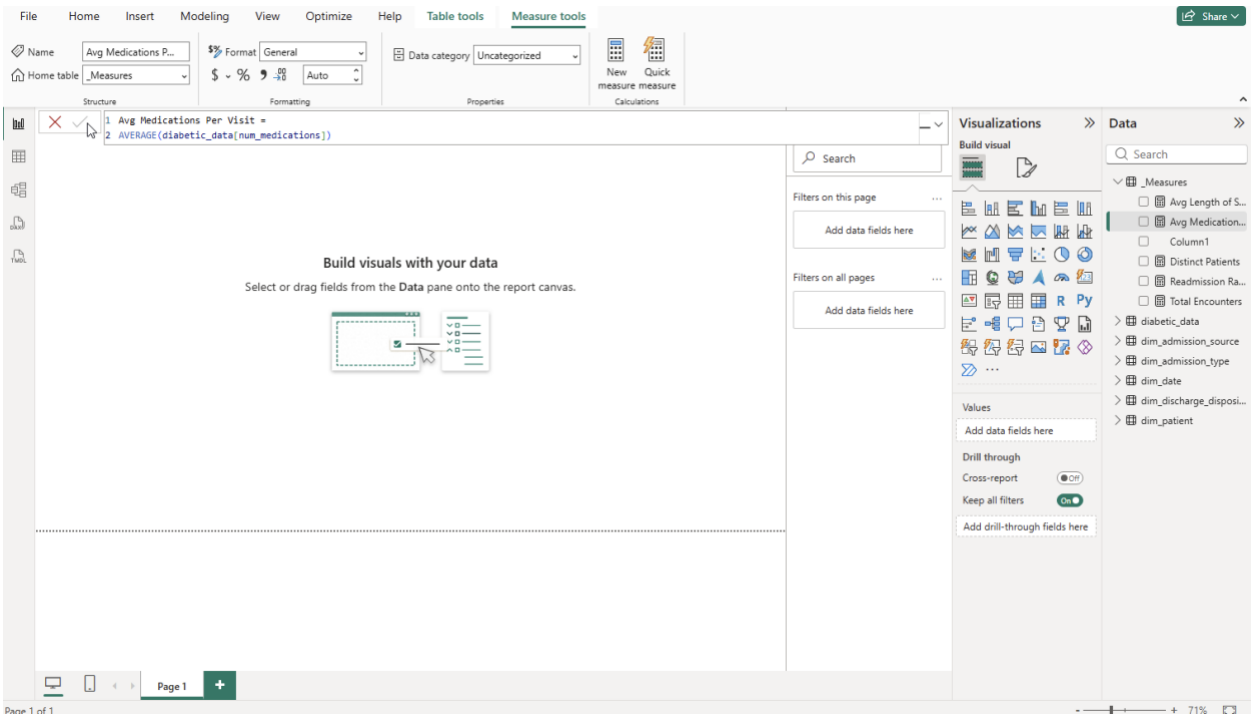
Primary operational efficiency metric. A high average length of stay signals either high clinical complexity or inefficient discharge processes.



M5 — Avg Medications Per Visit

```
Avg Medications Per Visit = AVERAGE(diabetic_data[num_medications])
```

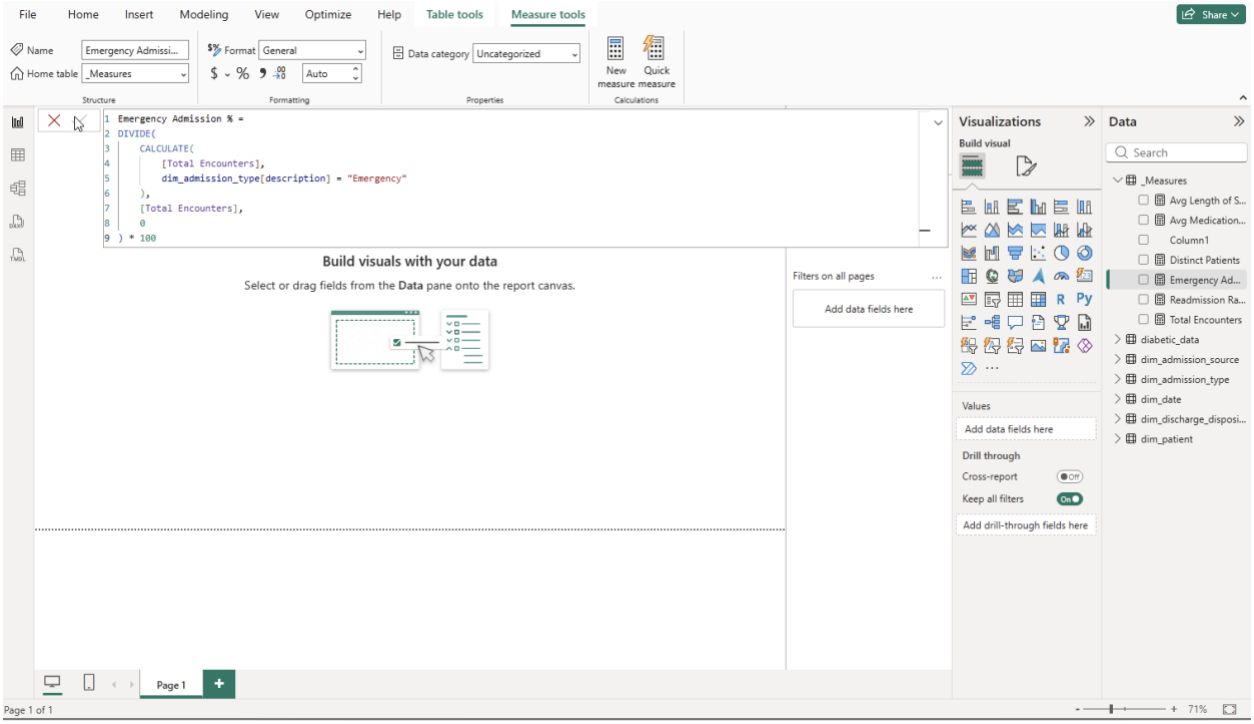
High medication counts indicate clinical complexity , which correlates strongly with longer hospital stays and higher readmission rates.



M6 — Emergency Admission %

```
Emergency Admission % =
DIVIDE(
    CALCULATE([Total Encounters],
        dim_admission_type[description] = "Emergency"),
    [Total Encounters], 0
) * 100
```

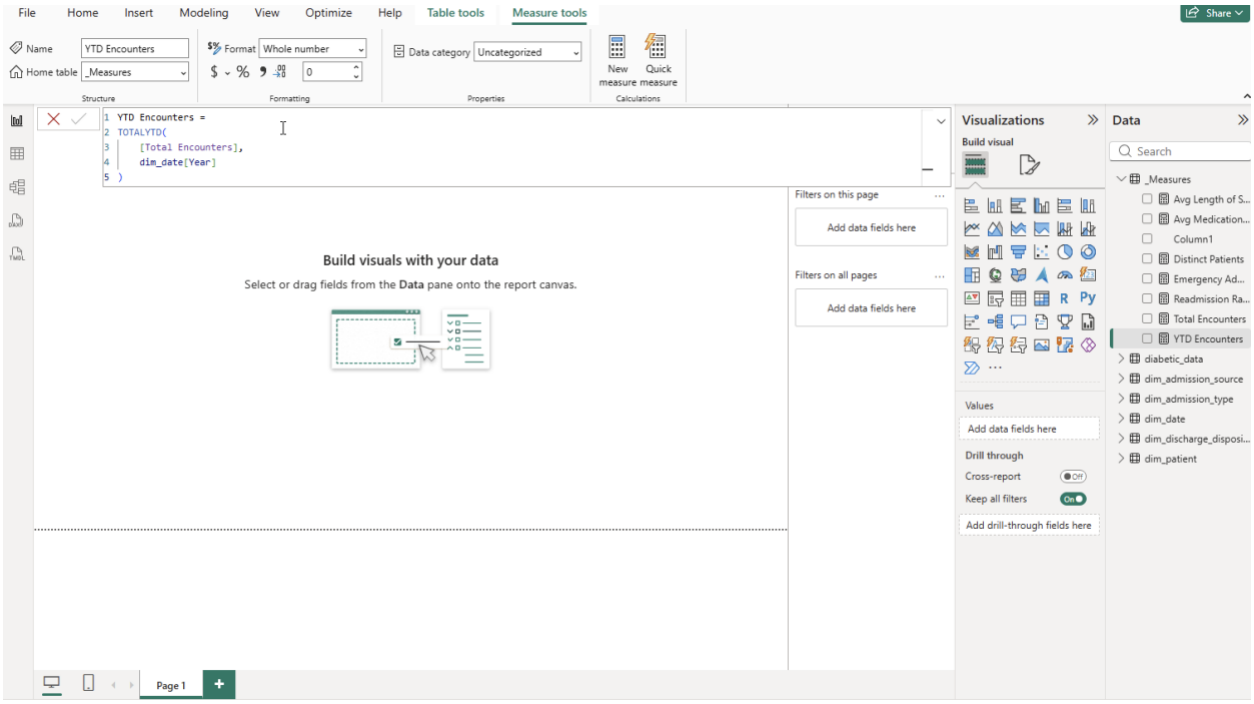
Measures what proportion of all encounters are unplanned emergency admissions. A high figure signals poor chronic disease management in the community.



M7 — YTD Encounters

```
YTD Encounters =
CALCULATE(
    [Total Encounters],
    FILTER(dim_date,
        dim_date[Year] <= MAX(dim_date[Year])
    )
)
```

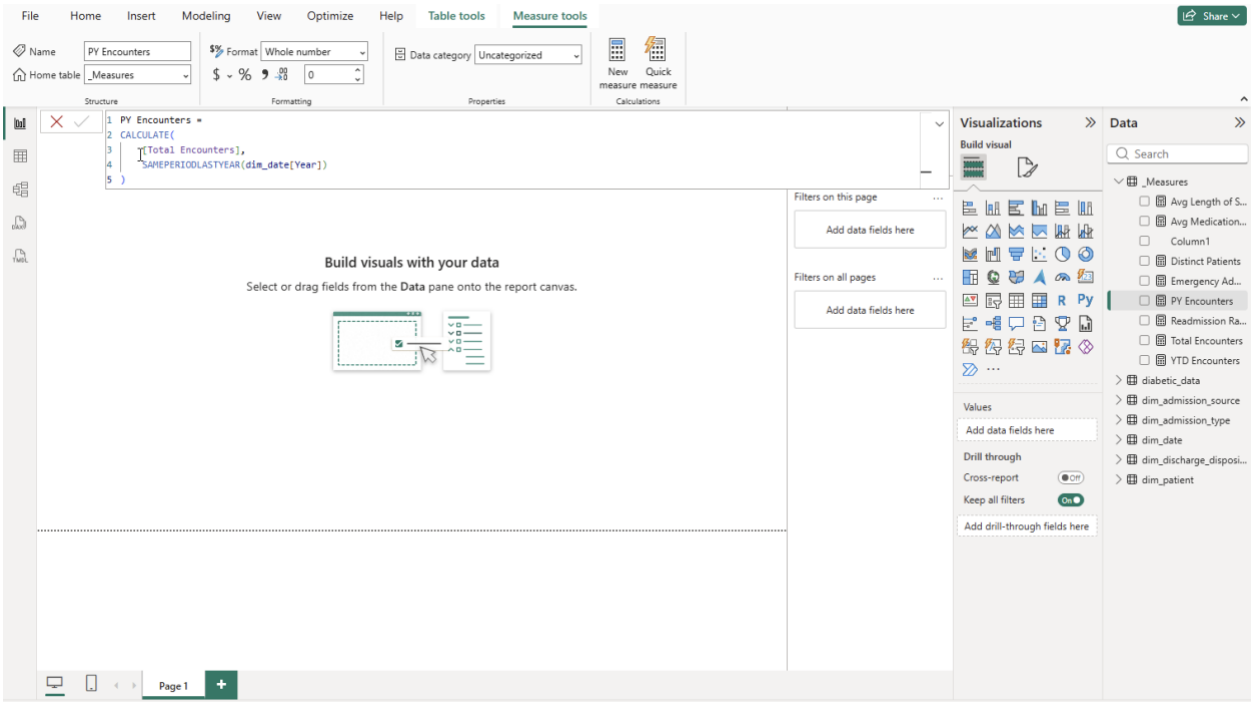
Accumulates total encounters from the start of the filtered year range. Used in time intelligence analysis on the Performance Monitoring page.



M8 — PY Encounters

```
PY Encounters =
CALCULATE(
    [Total Encounters],
    FILTER(dim_date,
        dim_date[Year] = MAX(dim_date[Year]) - 1
    )
)
```

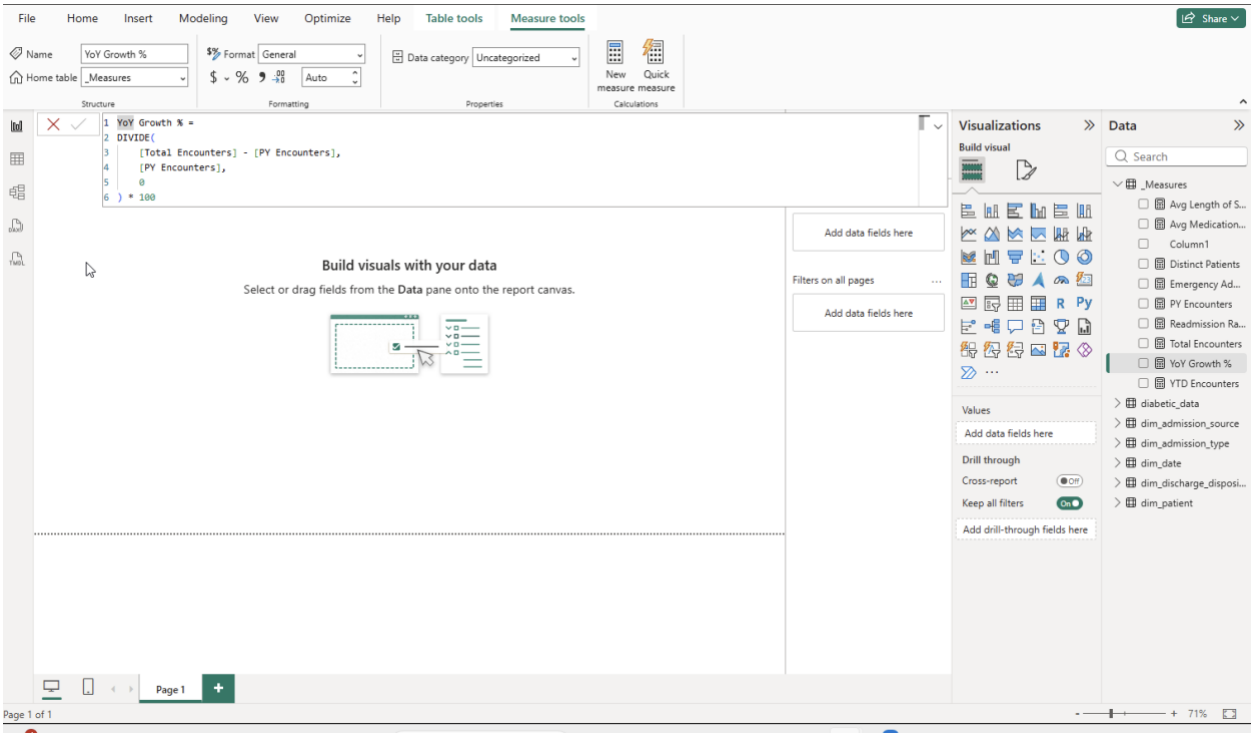
Returns total encounters for the prior year. Enables direct year-over-year comparison on trend charts alongside YTD Encounters.



M9 — YoY Growth %

```
YoY Growth % =
DIVIDE(
    [Total Encounters] - [PY Encounters],
    [PY Encounters], 0
) * 100
```

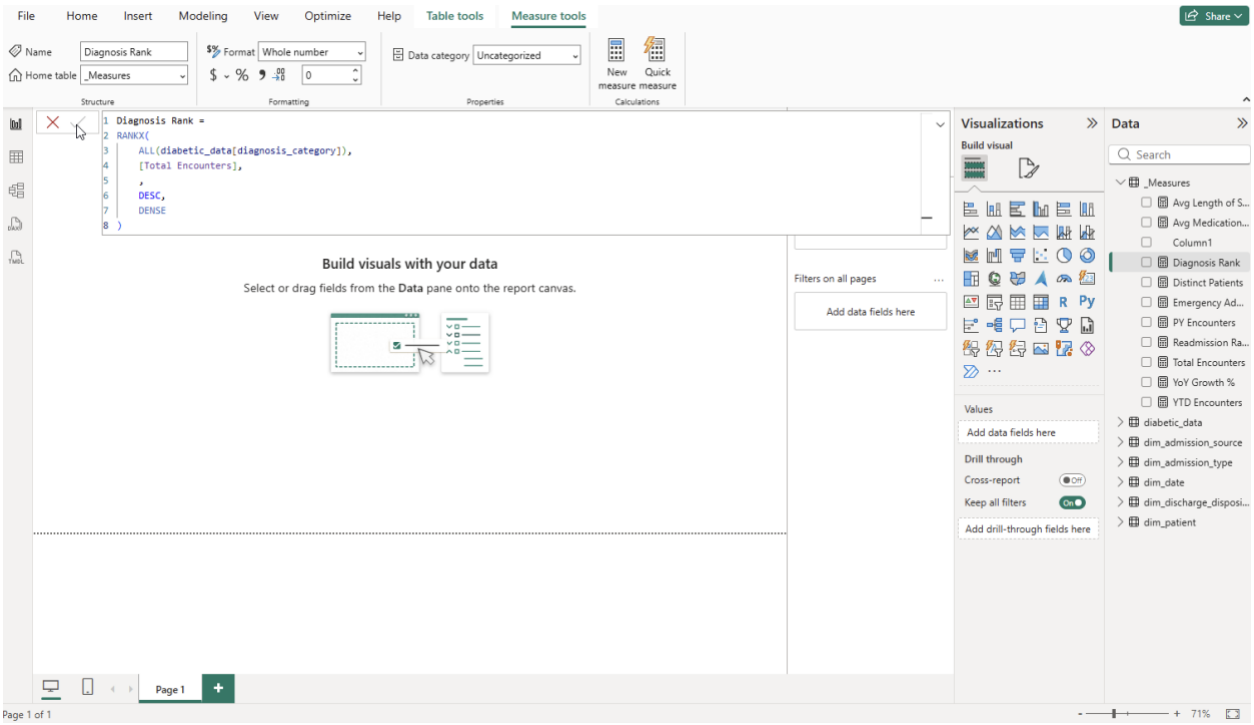
Shows year-over-year change in encounter volume. Positive values indicate growth; negative values may reflect improved preventive care reducing inpatient admissions.



M10 — Diagnosis Rank

```
Diagnosis Rank =
RANKX(
    ALL(diabetic_data[diagnosis_category]),
    [Total Encounters], , DESC, DENSE
)
```

Ranks diagnosis categories by encounter volume. ALL removes filter context so categories are ranked against each other regardless of dashboard selections.



PART E

Dashboard Design and Visualization

20 marks

I built a three-page interactive Power BI dashboard with a dark navy theme (#12181F page background, #1E2733 visual backgrounds). All pages share a consistent visual language: white typography, teal (#1D9E75) as the primary accent, coral (#E24B4A) for risk indicators, and blue (#378ADD) for informational elements.

Page 1 — Executive Summary

The Executive Summary page is designed to give any stakeholder an immediate high-level view of hospital encounter performance without requiring prior knowledge of the data. Four KPI cards run across the top showing Total Encounters, Distinct Patients, Readmission Rate %, and Average Length of Stay. These four figures answer the most critical operational and clinical questions at a glance.

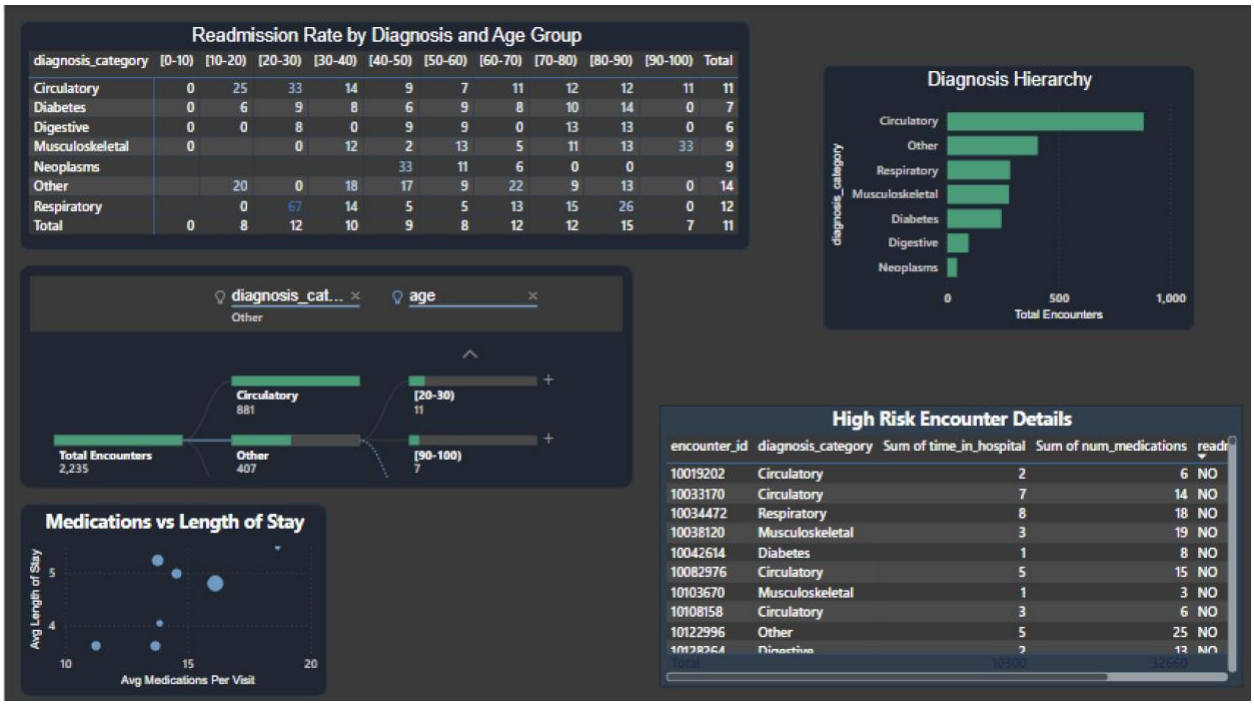
A year slicer with tile-style buttons runs across the full width at the top, allowing users to filter all visuals to a specific year instantly. Below the KPI cards, a line chart shows encounter volume by year, a horizontal bar chart shows encounters by age group, a donut chart breaks down admission

types, and a clustered bar chart shows 30-day readmission rate by patient race — surfacing health equity disparities across demographic groups.

Page 2 — Clinical Deep Dive

The Clinical Deep Dive page provides granular analysis for clinical and operational decision-makers. It moves beyond summary metrics to expose patterns within the data — which diagnoses drive the highest readmission rates across age groups, how medications relate to length of stay, and which individual encounters represent the highest clinical risk.

The decomposition tree visual is particularly powerful for interactive exploration, allowing users to click through layers of data to understand which combination of factors drives the highest encounter volumes. The scatter plot reveals the relationship between polypharmacy and hospital stay duration at a diagnosis category level, while the matrix provides a comprehensive cross-tabulation of readmission rates by diagnosis and age group with conditional formatting to highlight clinical hotspots.



Page 3 — Performance Monitoring

The Performance Monitoring page focuses on time-based analysis and performance benchmarking. It answers questions about how encounter volume and readmission rates have changed over time, which diagnosis categories consistently produce the highest readmission rates, and what proportion of admissions are unplanned emergency encounters.

The waterfall chart provides a compelling visual narrative of how encounter volumes changed year by year — green bars show years of growth and red bars show decline. The YTD and PY encounter line chart enables direct year-over-year comparison, while the readmission rate by year column chart identifies which years performed above or below the overall average of 11.2%.

PART F

Analytical Insights and Business Recommendations

10 marks

The following analysis was derived by examining patterns, anomalies, and trends visible across all three dashboard pages. These insights go beyond describing individual charts — they identify systemic issues, risks, and opportunities in the data that have direct implications for hospital management decisions.

Analytical insights

INSIGHT 1

Respiratory disease in young patients shows a disproportionately high readmission rate

The readmission matrix on Page 2 reveals that patients in the [20-30) age group with respiratory diagnoses show a readmission rate of 67%, which is more than six times the overall dataset average of 11.2%. This is a significant anomaly — young patients are generally expected to have lower readmission risk than elderly cohorts. The pattern suggests that respiratory conditions in

young diabetic patients may be poorly managed in outpatient settings, leading to a high rate of return within 30 days. This group represents a small volume but an outsized quality and cost risk.

INSIGHT 2

The 70-80 age group drives the highest encounter volume and carries above-average readmission risk simultaneously

The age group bar chart on Page 1 shows that the [70-80) cohort is the single largest patient group with 26,068 encounters. The readmission matrix on Page 2 shows this same cohort has readmission rates consistently above the average across multiple diagnosis categories, particularly in Circulatory (12%) and Musculoskeletal (11%) conditions. This dual burden of high volume and elevated risk places the greatest strain on inpatient bed capacity. Any intervention targeted at this age group will have the largest impact on overall hospital performance metrics.

INSIGHT 3

Emergency admissions represent over a third of all encounters, suggesting a chronic disease management gap

The Emergency Admission % KPI card on Page 3 shows 35.37% of all encounters are unplanned emergency admissions. For a chronic condition like diabetes, which is manageable in outpatient settings, this figure is high. Well-managed diabetic patients should rarely require emergency hospital care. This rate indicates that a substantial proportion of patients are not receiving adequate monitoring, medication management, or community health support between hospital visits, causing their conditions to deteriorate to the point of requiring emergency intervention.

INSIGHT 4

Encounter volume peaked around 2003 to 2004 and has been declining since, but readmission rates have not improved proportionally

The line chart and waterfall chart on Page 3 show that encounter volumes peaked around 2003 to 2004 and declined steadily through 2008. The column chart of readmission rates by year shows that despite lower overall encounter volumes in later years, the readmission rate remained at 10 to 13% throughout the period with no clear downward trend. This decoupling suggests that while the hospital network may have successfully reduced unnecessary admissions, it has not made equivalent progress in reducing preventable readmissions. Volume reduction and quality improvement require different interventions.

INSIGHT 5

The "Other" diagnosis category consistently records the highest readmission rate across all age groups

The readmission rate by diagnosis category bar chart on Page 3 and the matrix on Page 2 both show that encounters classified as "Other" — those whose primary ICD-9 code does not map to a standard clinical category — have the highest readmission rate in the dataset. This pattern likely

reflects diagnostic complexity and coding inconsistency. Patients presenting with atypical or multi-system conditions that do not fit standard ICD-9 categories may not receive targeted discharge planning, increasing their risk of returning within 30 days. This represents an opportunity to improve clinical coding and pathway management for complex cases.

Business recommendations

RECOMMENDATION 1

Implement a targeted post-discharge monitoring programme for young respiratory patients

Given the 67% readmission rate observed in the [20-30) age group with respiratory diagnoses, the hospital network should establish a structured 7-day and 14-day post-discharge follow-up protocol specifically for this cohort. This should include a phone consultation within 48 hours of discharge, a community respiratory nurse referral, and a scheduled outpatient visit within 7 days. Evidence from comparable programmes in chronic disease management suggests that structured follow-up reduces 30-day readmission rates by 20 to 30%. While this group is small in volume, the severity of the readmission rate warrants prioritised intervention.

RECOMMENDATION 2

Expand specialist geriatric outpatient capacity to address the 70-80 cohort burden

The [70-80) age group's combination of highest encounter volume and above-average readmission rates is creating a compounding strain on inpatient resources. The hospital network should invest in expanding specialist geriatric outpatient clinics with dedicated diabetes and circulatory disease management programmes targeting this cohort. Shifting follow-up care for these patients from inpatient to outpatient settings would reduce emergency readmissions, free inpatient beds for higher-acuity cases, and likely reduce the average length of stay for this group. A pilot programme of 3 months with clear readmission rate targets would allow evidence-based scaling of the initiative.

RECOMMENDATION 3

Introduce a comprehensive discharge planning protocol for patients with complex or unclassified diagnoses

The "Other" diagnosis category's consistently high readmission rate across all age groups indicates a systemic gap in discharge planning for clinically complex patients. The hospital network should introduce a mandatory multidisciplinary discharge review for all patients whose primary diagnosis falls outside the standard clinical coding categories. This review should include a pharmacist, a social worker, and the responsible clinician, with a focus on medication reconciliation, community support referrals, and a clear follow-up plan. Improving clinical coding

accuracy in parallel will also reduce the size of the "Other" category over time, making the patient population more analytically visible and clinically manageable.

PART G

GitHub Repository and Documentation

5 marks

Repository structure

```
DSA3050A-Advanced-PowerBI-Exam/
│
├── data/
│   ├── diabetic_data.csv
│   └── IDS_mapping.csv
│
├── screenshots/
│   ├── Dataset Source and Description/
│   ├── Data Transformation and Power Query/
│   ├── Data Modelling/
│   ├── DAX Measures/
│   └── Dashboard/
│
├── powerbi/
│   └── DSA3050A_Healthcare_BI.pbix
│
├── report/
│   └── DSA3050A_Nathan_Omenge_670637_Report.pdf
│
└── README.md
```

GITHUB REPOSITORY

https://github.com/Nathan-Omenge/DSA3050A-Advanced-PowerBI-Exam

Conclusion — Summary and Reflections

This project delivered a complete, professional Power BI analytical solution for diabetic patient encounter data across 130 US hospitals from 1999 to 2008. Every stage of the BI development process was executed from scratch — from raw data ingestion and Power Query transformation through relational data modeling, DAX measure development, and interactive dashboard design.

The analysis produced actionable insights with direct clinical implications: the unexpectedly high readmission rate in young respiratory patients, the dual burden of the 70-80 age cohort, and the persistent gap between declining encounter volumes and stagnant readmission rates. These findings provide the hospital network with a clear evidence base for where to target quality improvement investments.

The project also surfaced several technical challenges that required problem-solving beyond standard workflows — including the absence of a date column in the dataset, the mixed-format IDS_mapping source file, and the cardinality errors caused by duplicate ID values in the lookup tables. Each challenge was resolved and documented, demonstrating the real-world complexity of working with clinical data.

Nathan Orang'o Omenge · 670637 · DSA 3050A · April 2026

United States International University — Africa